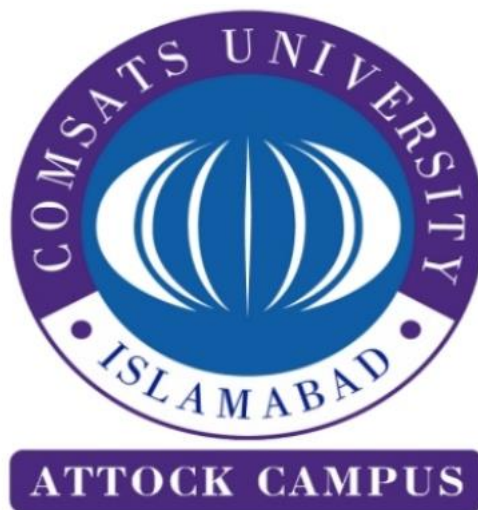


Semester: CE

Comsats University Islamabad Attock Campus



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Course Name: Programming Fundamentals

Program: Computer Engineering

Programming Fundamentals (Functions, Recursion, Fibonacci Sequence, and C Programming)

Question 1:

Write a function power (a, b) to calculate the value of a raised to b.

Program:

```
#include <stdio.h>
#include <stdlib.h>
int main()
{
    int a, b, res;
    printf("Enter value of number a: ");
    scanf("%d", &a);
    printf("Enter value of number b: ");
    scanf("%d", &b);
    res = power(a, b);
    printf("Result of a raised to b = %d\n", res);
}
int power(int a, int b)
{
    int x;
    x = pow(a, b);
    return x;
}
```

Output:

 "C:\Users\Hamza Computer\Documents\PF\lab1\quiz\bin\Debug\quiz.exe"

```
Enter value of number a: 3
Enter value of number b: 5
Result of a raised to b = 243
```

```
Process returned 30 (0x1E)   execution time : 6.132 s
Press any key to continue.
```

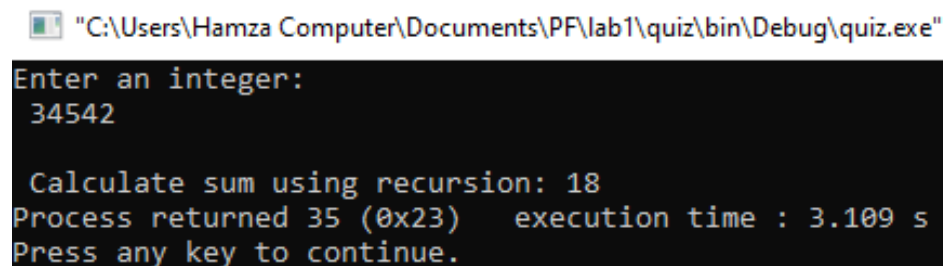
Question 2:

5-digit positive integer is entered through the keyboard, write a function to calculate sum of digits of the 5-digit number: (1) Without using recursion. (2) Using recursion.

Program:

```
#include <stdio.h>
#include <stdlib.h>
int Rec_func(int num);
int main()
{
    int num, Rec;
    printf("Enter an integer:\n ");
    scanf("%d", &num);
    Rec = Rec_func(num);
    printf("\n Calculate sum using recursion: %d",Rec);
}
int Rec_func(int num)
{
    if (num==0)
    {
        return 0;
    }
    return (num%10+Rec_func(num/10));
}
```

Output:



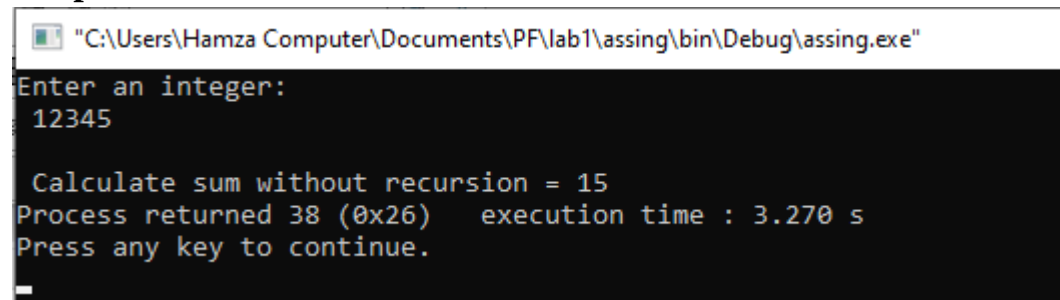
```
"C:\Users\Hamza Computer\Documents\PF\lab1\quiz\bin\Debug\quiz.exe"
Enter an integer:
34542

Calculate sum using recursion: 18
Process returned 35 (0x23)   execution time : 3.109 s
Press any key to continue.
```

Without Recursion:

```
#include <stdio.h>
#include <stdlib.h>
int Non_rec_func(int num);
int main()
{
    int num, Non_rec;
    printf("Enter an integer:\n ");
    scanf("%d", &num);
    Non_rec = Non_rec_func(num);
    printf("\n Calculate sum without recursion = %d", Non_rec);
}
int Non_rec_func(int num)
{
    int res, count=0;
    while(num!=0)
    {
        res=num%10;
        count=count+res;
        num = num/10;
    }
    return count;
}
```

Output:



```
"C:\Users\Hamza Computer\Documents\PF\lab1\assing\bin\Debug\assing.exe"
Enter an integer:
12345

Calculate sum without recursion = 15
Process returned 38 (0x26)   execution time : 3.270 s
Press any key to continue.
_
```

Question 3:

Write a program in C language to obtain the numbers of a Fibonacci sequence UPTIL THE USER REQUIRES. In a Fibonacci sequence the sum of two successive terms gives the third term. Following are the first few terms of the Fibonacci sequence: 1 1 2 3 5 8 13 21 34 55 89...

Program:

```
#include <stdio.h>
#include <stdlib.h>
int main()
{
    int num, First = 0, Second = 1, Next, i;
    printf("Enter the number of terms: ");
    scanf("%d", &num);
    printf("\nFibonacci series the sum of successive terms:\n ");
    for (i = 1; i <= num; i++)
    {
        printf("%d ", First);
        Next = First + Second;
        First = Second;
        Second = Next;
    }
    return 0;
}
```

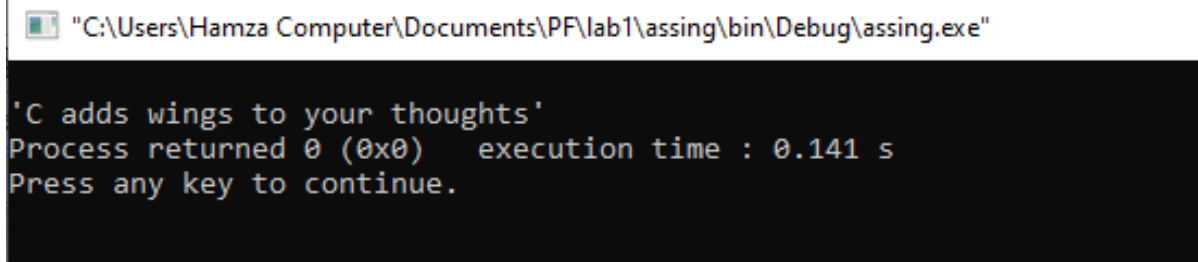
Output:

Question 4: What would be the output of the following programs:

(a) Program:

```
main()  
{  
    int i = 0 ;  
    i++ ;  
    if ( i <= 5 )  
    {  
        printf ( "\nC adds wings to your thoughts" ) ;  
        exit( ) ;  
        main( ) ;  
    }  
}
```

Output:



The screenshot shows a Windows command prompt window with the title bar "C:\Users\Hamza Computer\Documents\PF\lab1\assing\bin\Debug\assing.exe". The output of the program is displayed in the command prompt:

```
'C adds wings to your thoughts'  
Process returned 0 (0x0)   execution time : 0.141 s  
Press any key to continue.
```

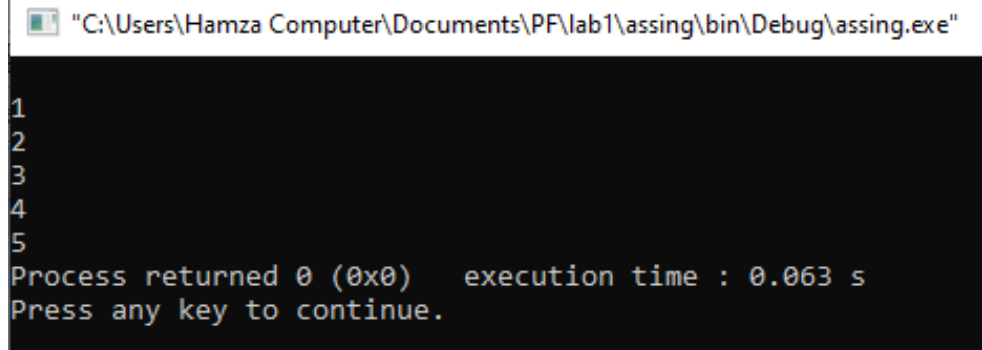
This program will print the message **“C adds wings to your thoughts”** once and then terminate, without printing anything else. This is because the variable *i* is incremented to 1, and then the if statement checks if *i* is less than or equal to 5, which is true. The printf statement is executed, and then the program exits using the `exit()` function, which terminates the program immediately and does not allow any further code to execute.

(b) Program:

```
Program:main()  
{  
    static int i = 0 ;  
    i++ ;  
    if ( i <= 5 )  
    {  
        printf ( "\n%d", i ) ;  
        main( ) ;  
    }  
}
```

```
else exit( ) ;  
}
```

Output:



```
"C:\Users\Hamza Computer\Documents\PF\lab1\assing\bin\Debug\assing.exe"  
1  
2  
3  
4  
5  
Process returned 0 (0x0) execution time : 0.063 s  
Press any key to continue.
```

This program will print the **numbers 1 through 5, one per line**, and then terminate. This is because the variable 'i' is declared as a static variable, which means that it retains its value between function calls. The first time the function is called, 'i' is initialized to 0 and then incremented to 1. The if statement checks if I is less than or equal to 5, which is true, so the current value of I (which is 1) is printed to the console using printf. The function is then called recursively using the main () function, and I is incremented again to 2. This process continues until I reaches 6, at which point the if statement condition is false, and the program terminates using the exit () function.